

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

*I D.Anand Paul (192425207)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

D.Anand Paul.

Signature of the student:

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Scenario: Event Communication Security

A student organization wants to share confidential information about an upcoming event with its core members. To protect the message from unauthorized people, they use a simple classical encryption technique called the Caesar Cipher. In this method, every letter of the message is shifted by a fixed number of positions in the alphabet.

The given message is:

Original Message: MEET AT NOON

The shift value is **4 positions**. Therefore, each letter is moved four positions forward in the alphabet.

For example:

- $M \rightarrow Q$
- $E \rightarrow I$
- $E \rightarrow I$
- $T \rightarrow X$
- $A \rightarrow E$
- $T \rightarrow X$
- $N \rightarrow R$
- $O \rightarrow S$
- $O \rightarrow S$
- $N \rightarrow R$

Therefore, the encrypted message becomes:

Encrypted Message: QIIX EX RSSR

The spaces between the words can be kept unchanged to make the message easier to read.

An authorized member who knows the encryption rule can easily retrieve the original message.

The member uses the **decryption process**, where every encrypted letter is shifted **4 positions backward**.

For example:

- $Q \rightarrow M$
- $I \rightarrow E$
- $I \rightarrow E$
- $X \rightarrow T$
- $E \rightarrow A$
- $X \rightarrow T$
- $R \rightarrow N$
- $S \rightarrow O$
- $S \rightarrow O$
- $R \rightarrow N$

Thus:

QIIX EX RSSR → MEET AT NOON

The Caesar Cipher is simple and easy to understand, but it provides only basic security because the number of possible shifts is very small. An attacker can try all possible shifts and discover the original message.

Conclusion:

The Caesar Cipher can be used to demonstrate basic encryption and decryption. In this scenario, a shift of 4 changes “MEET AT NOON” into “QIIX EX RSSR”, and an authorized person can decrypt it using the same shift value in the reverse direction.

Scenario: Suspicious Digital Asset Investigation

During a digital forensic investigation, investigators suspect that a company logo contains hidden confidential information. Although the image looks normal to the human eye, secret information may have been embedded inside the image using a technique called **steganography**.

Steganography is a method of hiding information inside another file, such as an image, audio file, video, or document. The main purpose is to make the existence of the hidden information difficult to notice. Investigators can begin by examining the **file properties and metadata** of the image. They can check the file size, image dimensions, file format, creation details, and other metadata. An unusually large file size compared with a similar normal logo may indicate the presence of hidden information.

Another important method is **pixel analysis**. Digital images are made up of pixels containing colour values. Secret information can be hidden by making very small changes to these values. Investigators can analyze the **Least Significant Bits (LSB)** of the pixels to identify unusual patterns that may contain hidden data. Investigators can also use **steganography detection or forensic tools** to analyze the image. These tools can identify unusual patterns, extra data, or information appended to the image file. Comparing the suspicious logo with an original or clean copy can also help identify differences.

Two techniques can be used to strengthen the security of hidden communication:

1. Encryption before hiding:

The confidential message should first be encrypted using a strong encryption algorithm. The encrypted message can then be hidden inside the image. Even if investigators extract the hidden data, they cannot easily understand it without the encryption key.

2. Randomized embedding:

Instead of hiding information in a simple or predictable sequence of pixels, the data can be distributed across randomly selected pixels using a secret key. This makes statistical detection and extraction more difficult.

Therefore, investigators can combine metadata analysis, pixel analysis, file comparison, and forensic tools to detect hidden information.

Conclusion:

Steganography can make confidential communication appear like an ordinary image. However, careful forensic analysis can reveal unusual patterns or hidden data. Encryption and randomized embedding can provide additional protection for hidden communications.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation